

Souhaieb Elamri

Electrical Engineering Student | Computer Vision and Industrial AI for Automation

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github.com/souhaieb-elamri | souhaieb-elamri.github.io/Portfolio-

PROFILE

Electrical Engineering student at ENSIT with a Master's in Advanced Robotics and Artificial Intelligence. Substantial hands-on experience in computer vision, with strong technical skills in embedded systems, industrial automation, and Industrial AI. Seeking a final-year engineering internship starting February 2027.

PROFESSIONAL EXPERIENCE

Summer Intern – WIC MIC GROUP

06/2026 – 08/2026 | Bizerte, Tunisia

Tools: Python, YOLO11, YOLO11-Pose, DocTR, SVM, CatVTON, OpenCV, FastAPI, Streamlit, WebRTC, Docker, SQL Server

- Designed and deployed an AI-powered production ticket digitization system using YOLO11, SVM, FastAPI, and SQL Server, automating extraction and storage of production data while eliminating manual transcription.
- Developed an AI-based virtual garment try-on and body measurement platform using CatVTON, YOLO11-Pose, ArUco calibration, Streamlit, and WebRTC to support garment visualization and centimeter-level body measurements for industrial applications.

Master's Thesis Intern – WIC MIC GROUP

02/2025 – 05/2025 | Bizerte, Tunisia

Tools: Python, OpenCV, PyTorch, YOLO11-Pose, CUDA, Docker, SQL Server

- Engineered a real-time AI garment inspection system, managing everything from massive data annotation to OpenCV multi-camera calibration, achieving ± 0.3 cm precision.

Bachelor's Project Intern – HYCO Industries – Metal Construction

02/2023 – 05/2023 | Nabeul, Tunisia

- Retrofitted a manual metal fabrication machine into an automated multi-axis CNC system, integrating stepper motor drivers and configuring Mach3 control software via a Raspberry Pi interface.

Technical Intern – CMR Tunisia Ship Repairs

01/2023 – 02/2023 | Bizerte, Tunisia

- Automated a marine basin control system by developing a PLC/HMI architecture using Siemens PLCs, Ladder Logic, and TIA Portal for real-time pump, valve, and water-level control.

EDUCATION

Engineering Degree in Electrical Engineering

2025 – 2027 (expected)

National Higher School of Engineers of Tunis (ENSIT)

Master's Degree in Advanced Robotics and Artificial Intelligence

2023 – 2025

Higher Institute of Technological Studies of Bizerte

Bachelor's Degree in Electrical Engineering – Industrial Automation

2020 – 2023

Higher Institute of Technological Studies of Nabeul

ACADEMIC PROJECTS

Smart Embedded System for Wildfire Prevention

2026 – Present

- Developing a Raspberry Pi-based system that reads soil-dryness, CO2, wind and temperature sensors, transmits data with link-status messages, and flags conditions with a risk of fire.

Intelligent Mobile Robot

2025 – 2026

- Designed and developed a Bluetooth-controlled mobile robot with PWM motor control, interrupt-based emergency stop, real-time sensor acquisition (ADC, I2C), and IoT monitoring via ESP8266, ThingSpeak, and Node-RED.

TECHNICAL SKILLS

Embedded & Industrial: Raspberry Pi, Arduino, ESP32, STM32, FPGA, VHDL, TIA Portal, PLC, CNC, LabVIEW

AI & Computer Vision: TensorFlow, PyTorch, OpenCV, MediaPipe, YOLO, Object Detection, Pose Estimation, OCR

Tools & Languages: Python, C, MATLAB/Simulink, SolidWorks, FastAPI, SQL Server, Docker, Git

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900)  – 05/2024

Microsoft Certified: Security, Compliance, and Identity Fundamentals (SC-900)  – 10/2024

Six Sigma Yellow Belt  – 01/2025

LANGUAGES

French – Fluent | **English** – Fluent | **Arabic** – Native